

Cryptography Notes

Oli

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1. Algebra

1.1. Notation

- $\mathbb{Z}$ = 整數 integers
- $\mathbb{Z}_n = \{0, 1, 2, \dots, n-1\}$ ($0 \sim n-1$ 的整數)
- $\mathbb{Z}_n^* = \{a \in \mathbb{Z}_n \mid \gcd(a, n) = 1\}$ ($\mathbb{Z}_n$ 裡面和 n 互質的)

e.g. $\mathbb{Z}_{12}^* = \{1, 5, 7, 11\}$

- $\mathbb{N}$ = non-negative integers (有 0)
- $\mathbb{P}$ = positive integers (無 0)
- $\mathbb{Q}$ = rational numbers
- $\mathbb{R}$ = real numbers
- $\mathbb{C}$ = complex numbers
- $\mathbb{Q}^*, \mathbb{R}^*, \mathbb{C}^* = \text{non-zero rational, real, complex numbers}$
- $GL_n(\mathbb{R})$
 - $n \times n$ 的可逆 (invertible) 方陣
 - 裡面的元素 (entries) 都在 $\mathbb{R}$ 裡
- $SL_n(\mathbb{R})$
 - $n \times n$ det 為 1 的方陣 (det 為 1 所以也可逆)
 - 裡面的元素都在 $\mathbb{R}$ 裡, 是 $GL_n(\mathbb{R})$ 的 subset

1.2. Group

1.2.1. Definition

group $(G, *)$ 是一個集合 G 跟一個運算 $*$ 所構成的 (中文是群)

一個 group 要滿足四個條件：

1. Closure 封閉性

$$a * b \in G \quad \forall a, b \in G$$

$a * b$ 後的答案還是得在 G 裡面

2. Associativity 結合律

$$a * (b * c) = (a * b) * c \quad \forall a, b, c \in G$$

3. Identity

$$\exists e \in G \text{ s.t. } a = a * e = e * a$$

找得到一個 identity e 讓 a 和 e 運算或 e 和 a 運算後還是自己

* 通常是加或乘

加法的 identity e 通常是 0 (0 加自己或自己加 0 都還是自己)

減法的 identity e 通常是 1 (1 乘自己或自己乘 1 都還是自己)

4. Inverse 反元素

$$\forall a \in G, \exists b \in G \text{ s.t. } a * b = b * a = 1$$

找得到跟和 a 運算完會變 identity 的 inverse b

Group 的 Example

- $\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$ with $+$
- $\mathbb{Q}^*, \mathbb{R}^*, \mathbb{C}^*$ with $\times$
- $5\mathbb{Z} = \{5a \mid a \in \mathbb{Z}\}$ with $+$
- $\{1, -1\}$ with $\times$
- $\mathbb{Z}_6 = \{0, 1, 2, 3, 4, 5\}$ with $+\text{mod } 6$
- $\mathbb{Z}_7^*$ with $\otimes$ ($\otimes$ 表示 $\times \text{mod } p$, p 是 $\mathbb{Z}$ 的下標 所以在這邊 $\otimes = \times \text{mod } 7$)
- $\{(x, y) \in \mathbb{R}^2 \mid y^2 = x^3 + ax + b\} \cup \{\infty\}$ with point addition (加法運算：兩個點相加) and point doubling (兩倍運算：一個點的兩倍) on elliptic curves

不符合 Group 的 Example

- $\mathbb{Z}_7$ with $+$ $\Rightarrow$ not closed
- $\mathbb{P}$ with $+$ $\Rightarrow$ no identity
- $\mathbb{Z}$ with $-$ $\Rightarrow$ no associativity
- $2\mathbb{Z} + 1 = \{2a + 1 \mid a \in \mathbb{Z}\}$ (奇數) with $\times$ $\Rightarrow$ no inverse

1.2.2. Proposition

1. The identity e of G is unique

Proof

假設有兩個 identity e_1 跟 $e_2 \Rightarrow e_1 = e_1 * e_2 = e_2$

2. The inverse b of each $a \in G$ are unique, denoted as a^{-1}

Proof

假設 b 跟 c 都是 a 的 inverse $\Rightarrow b = b * e = b * (a * c) = (b * a) * c = e * c = c$

3. $(a^{-1})^{-1} = a \forall a \in G$
4. $(a * b)^{-1} = b^{-1} * a^{-1} \forall a, b \in G$
5. $\forall a, b, c \in G$
 - a. $a * x = b$ and $y * a = b$ have unique solution

Proof

$a^{-1} * (a * x) = a^{-1} * b \rightarrow (a^{-1} * a) * x = a^{-1} * b \rightarrow e * x = a^{-1} * b$
 $\Rightarrow x = a^{-1} * b$

- b. $a * b = a * c$ 代表 $b = c$ and $a * b = c * b$ 代表 $a = c$

Proof

兩邊同乘 a^{-1} 然後套一下四個性質

1.2.3. Permutation

A **permutation** is a **bijective** function $f : T \rightarrow T$ on a set T into itself

injective 單射 (one-to-one)	surjective 滿射 (onto)	bijective 雙射 (one-to-one and onto)

Permutation 的 Example

$T = \{1, 2, 3\}$ with $f(1) = 2, f(2) = 3, f(3) = 1$ is a permutation represented by the array

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{bmatrix}$$

就是矩陣裡面上面那排的經過 f 函數會被換成下面那排的

Remark

Every block cipher with a specific key is a permutation

- $DES_k : \{0, 1\}^{64} \rightarrow \{0, 1\}^{64}$ (block size = 64 bits)
- $AES_k : \{0, 1\}^{128} \rightarrow \{0, 1\}^{128}$ (block size = 128 bits)

1.2.4. Symmetric Group

A **symmetric group** S_n is the set of all permutations on $\{1, 2, 3, \dots, n\}$ with $\circ$ (composition)

$(f \circ g)(x) = f(g(x))$ (合成函數 先運算後面的再運算前面的)

Symmetric Group 的 Example

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{bmatrix} \circ \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{bmatrix}$$

$$S_3 = \left\{ \begin{bmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{bmatrix}, \begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{bmatrix}, \begin{bmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{bmatrix}, \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{bmatrix} \right\}$$

可以把上面那些東西簡寫 : $S_3 = \{e, (12), (13), (23), (123), (132)\}$

- $(123)^{-1} = (321) = (132)$

Remark

The S-box of AES is a permutation in S_{256}

1.2.5. Abelian Group

A group $(G, *)$ is **commutative** or **abelian** if $a * b = b * a \forall a, b \in G$

Abelian Group 的 Example

- $\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$ with $+$
- $\mathbb{Z}_9^* = \{1, 2, 4, 5, 7, 8\}$ with $\otimes$
在 $\mathbb{Z}_9^*$ 中 2 和 5 的次方都會跑遍所有元素 所以他們是 $\mathbb{Z}_9^*$ 的生成元 (generator)
以 2 為例: $1 = 2^6, 2 = 2^1, 4 = 2^2, 5 = 2^5, 7 = 2^4, 8 = 2^3$
- $\mathbb{Z}_p^*$ with $\otimes$ for every prime p

Non-Abelian Group 的 Example

只要找到任意一個 $a * b \neq b * a$ 就是了

- for $n \geq 3$, the symmetric group S_n is not abelian :

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{bmatrix} \circ \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{bmatrix} \neq \begin{bmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{bmatrix} \circ \begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{bmatrix}$$

- with the operation of matrix multiplication, each of the following is not abelian :
 - ▶ A general linear group $GL_n(\mathbb{R})$
 - ▶ A special linear group $SL_n(\mathbb{R})$

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} \neq \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

the above is in $SL_2(\mathbb{Q}) \subset GL_2(\mathbb{Q})$

1.2.6. Cyclic Group

A group $(G, *)$ is **cyclic** if there exists a generator $g \in G$ s.t. every $a \in G$ is of the form $a = g * g * g * \dots * g$ (n copies) for some $n \in \mathbb{Z}$ (n 可以 < 0)

- 2 copies $\Rightarrow a = g * g$
- -2 copies $\Rightarrow a = g^{-1} * g^{-1}$

Cyclic Group 的 Example

- $(\mathbb{Z}, +)$ with generators 1 and -1
- $(\mathbb{Z}_7^*, \otimes)$ with generator 3
- $(\mathbb{Z}_9^*, \otimes)$ with generators 2 and 5

不是 Cyclic Group 的 Example

- $(\mathbb{Q}, +)$
- $(\mathbb{Z}_8^*, \otimes)$ (這種有 4 個元素而且都不是 generator 的 group 叫 Klein 4)

1.2.7. Group Order

The **order** of a group $(G, *)$ (denoted as $|G|$ or $\#G$) is the number of the elements in G , or the **cardinality** of G

Group Order 的 Example

- $|S_3| = 6, |S_4| = 24, |S_n| = n!$
- $|\mathbb{Z}_p| = p, |\mathbb{Z}_p^*| = p - 1$ for any prime p
- $|\mathbb{Z}_9^*| = 6$
- $|\mathbb{Z}_n^*| = |\{a \in \mathbb{Z}_n \mid \gcd(a, n) = 1\}| = \phi(n)$
 $\phi(n)$ 叫 euler phi-function
- 用來 construct AES 的 S-box 的 $|GL_8(\mathbb{Z}_2)| = ??$

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 & 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 0 & 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 1 & 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 & 1 \end{bmatrix} \in GL_8(\mathbb{Z}_2) (= SL_8(\mathbb{Z}_2))$$

A group $(G, *)$ is **finite** if $|G| < \infty$, and is **infinite** if $|G| = \infty$

Finite / Infinite 的 Example

- $|\mathbb{Z}| = \infty, |\mathbb{Q}| = \infty, |\mathbb{R}| = \infty, |\mathbb{C}| = \infty$
 - $|\mathbb{Z}|$ and $|\mathbb{Q}|$ have the same cardinality, which is countable
 - $|\mathbb{R}|$ and $|\mathbb{C}|$ have the same cardinality, which is uncountable

1.2.8. Subgroup and Coset

A subset H of a group $(G, *)$ is a **subgroup** of G if H is itself a group under $*$ in G

A subgroup H of a group $(G, *)$ is a subgroup of G iff

1. $a * b \in H \forall a, b \in H$
2. $a^{-1} \in H \forall a \in H$

Subgroup 的 Example

- $(5\mathbb{Z}, +)$ is a subgroup of $(\mathbb{Z}, +)$
- $(\mathbb{Q}^*, \times)$ is a subgroup of $(\mathbb{R}^*, \times)$

If H is a subgroup of a group G , and $g \in G$, then

- $gH := \{g * h \mid h \in H\}$ is a **left coset of H** in G
- $Hg := \{h * g \mid h \in H\}$ is a **right coset of H** in G
- for abelian groups, $gH = Hg$

Remark $\Rightarrow$ In general, Left coset $\neq$ Right coset

☰ Coset 的 Example

- $(123)\{e, (12)\} = \{(123), (13)\}$ ((123) 是 S_3 的元素, $\{e, (12)\}$ 是 S_3 的 subgroup)
 $\{e, (12)\}(123) = \{(123), (23)\}$
 $\Rightarrow L \neq R$
- $(12)\{e, (123), (132)\} = \{(12), (23), (13)\}$
 $\{e, (123), (132)\}(12) = \{(12), (13), (23)\}$
 $\Rightarrow L = R$

$$g_1 H = g_2 H \Leftrightarrow g_1^{-1} g_2 \in H$$

e.g. $(123)\{e, (12)\} = (13)\{e, (12)\} \Leftrightarrow (123)^{-1}(13) = (132)(13) = (12) \in \{e, (12)\}$

1.2.9. Lagrange Theorem

If H is a subgroup of a finite group G , then $|H|$ divides $|G|$

H is a subgroup of a finite group G , then $[G : H] := \frac{|G|}{|H|}$ is the **index** of H in G

For $g \in G$, the **order** of g , denoted by $o(g)$, is the smallest positive integer n s.t. $g^n = e$
 $\rightarrow o(g)$ divides $|G| : \langle g \rangle = \{g, g^2, g^3, \dots, g^{o(g)-1}, g^{o(g)} (= e)\}$ is a subgroup of G

$\langle \rangle$ 用在 group 時代表 subgroup, 用在 ring 時代表 ideal

☰ 這個性質的 Example

- $\{e, (123), (132)\}$ is a subgroup of S_3 , $3 \mid 6$
- $\{1, 8\}$ is a subgroup of $(\mathbb{Z}_9^*, \otimes)$, $2 \mid 6$
- $SL_2(\mathbb{Z}_7)$ is a subgroup of $GL_2(\mathbb{Z}_7)$
 $|GL_2(\mathbb{Z}_7)| = (7^2 - 1)(7^2 - 7)$, $|SL_2(\mathbb{Z}_7)| = \frac{|GL_2(\mathbb{Z}_7)|}{6}$ (下面有解釋)
- $T_2(\mathbb{Z}_7) = \{M \in GL_2(\mathbb{Z}_7) \mid M \text{ is upper triangular}\}$ is a subgroup of $GL_2(\mathbb{Z}_7)$
 $|T_2(\mathbb{Z}_7)| = 6 \times 6 \times 7$

$|SL_2(\mathbb{Z}_7)| = \frac{|GL_2(\mathbb{Z}_7)|}{6}$ 的解釋 (可以推廣至 $|SL_n(\mathbb{Z}_p)|, |GL_n(\mathbb{Z}_p)|$)

1. Let $H = SL_2(\mathbb{Z}_7) = \left\{ \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \mid a_{ij} \in \mathbb{Z}_7, a_{11}a_{22} - a_{12}a_{21} = 1 \right\}$
2. For $t = 2, 3, 4, 5, 6$, define the left coset $H_t = \begin{bmatrix} 1 & 0 \\ 0 & t \end{bmatrix} H$, then
 - a. $H_t = \{M \in GL_2(\mathbb{Z}_7) \mid \det(M) = t\}$
 - b. H and H_t 's are mutually disjoint (兩兩互斥: 任兩個集合中沒有共同元素)
 - c. $f_t : H \rightarrow H_t$ defined by $f_t(M) = \begin{bmatrix} 1 & 0 \\ 0 & t \end{bmatrix} M$ is bijective
 - d. $|H| = |H_t|$
 - e. $GL_2(\mathbb{Z}_7) = H \cup H_2 \cup H_3 \cup H_4 \cup H_5 \cup H_6$
3. $|GL_2(\mathbb{Z}_7)| = |H| + |H_2| + |H_3| + |H_4| + |H_5| + |H_6| = 6 \times |SL_2(\mathbb{Z}_7)|$

1.2.10. Homomorphism and Isomorphism

$(G, *)$ and $(H, \star)$ are two groups

- A function $f : G \rightarrow H$ is a **homomorphism** (同態) if
 - $f(a * b) = f(a) \star f(b) \forall a, b \in G$
- A function $f : G \rightarrow H$ is a **isomorphism** (同構) (G is isomorphic to $H / G \cong H$) if
 - $f(a * b) = f(a) \star f(b) \forall a, b \in G$
 - f is bijective

Homomorphism 的 Example

- $f : (\mathbb{Z}, +) \rightarrow (\mathbb{Z}, +)$ is a homomorphism with $f(1) = 2$
 - injective (one-to-one), not surjective (onto)
- $f : (\mathbb{Z}, +) \rightarrow (\mathbb{Z}_7^*, \otimes)$ is a homomorphism with $f(1) = 3$
 - surjective (onto), not injective (one-to-one)

Isomorphism 的 Example

- $f : (\mathbb{C}^*, \times) \rightarrow (\mathbb{C}^*, \times)$ defined by $f(a + bi) = a - bi$
- $f : (\mathbb{Z}_6, + \bmod 6) \rightarrow (\mathbb{Z}_7^*, \times \bmod 7)$ is a isomorphism defined with $f(1) = 3$
 - $f(2) = f(1 + 1) = 2, f(3) = f(1 + 1 + 1) = 6, f(4) = 4, f(5) = 5, f(0) = 1$
 - $f(0) = 1$ 因為 $f(0) = f(0 + 0) = f(0) \times f(0)$ 兩邊同乘 $f(0)^{-1}$ 得到 $f(0) = 1$
 - 思考：如果 $f(1) = 5, f(1) = 2$ 會怎麼樣

1.3. Ring

1.3.1. Definition

A **ring** $(R, +, \times)$ is a set R with two binary operations $+$ and $\times$ s.t. :

1. $(R, +)$ is an Abelian group
2. Closed under $\times : a \times b \in R$ for all $a, b \in R$
3. Associativity under $\times : a \times (b \times c) = (a \times b) \times c$ for all $a, b, c \in R$
4. Distributive laws :
 - $a \times (b + c) = a \times b + a \times c$
 - $(a + b) \times c = a \times c + b \times c$

for all $a, b, c \in R$

Ring 的 Example

- $\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}, n\mathbb{Z} = \{nt \mid t \in \mathbb{Z}\}$ with usual operations
- $\mathbb{Z}_n = \{0, 1, 2, \dots, n-1\}$ with $+, \times \bmod n$
- $C(\mathbb{R})$: continuous functions on $\mathbb{R}$ (單位元是 0 函數, $(f + g)(x) := f(x) + g(x)$)
- $R[x]$: polynomials with usual operations (係數都在 R 裡面)
 - $\{a_0 + a_1x + a_2x^2 + \dots + a_nx^n \mid a_i \in R, a_n \neq 0, n \in \mathbb{Z}^+\}$
- $R[x, y] = R[x][y]$ ($R[x]$ 是上一項 R 的角色, 也是一個在 R 裡面的係數)
 - a polynomial ring with two indeterminates, can be generalize to any amount

A ring $(R, +, \times)$ is called a **ring with identity** if there exists $1 \in R$ s.t. $a \times 1 = 1 \times a = a$ for all $a \in R$

A ring $(R, +, \times)$ is commutative if $a \times b = b \times a$ for all $a, b \in R$

NO identity 還有 NOT commutative 的 Example

- $M_n(\mathbb{R})$ ($n \times n$ matrices with usual operations) is not commutative
- $2\mathbb{Z}$ has no identity

1.3.2. Proposition

1. $a \times 0 = 0 \times a$ for $a \in R$

Proof

$$a \times 0 + a \times 0 = a \times (0 + 0) = a \times 0$$

$\Rightarrow$ 兩邊同時加上加法反元素就可以得到 $a \times 0 = 0$

2. $(-a) \times b = a \times (-b) = -(a \times b)$

Proof $(-a) \times b = -(a \times b)$

$$(-a) \times b + a \times b = [(-a) + a] \times b = 0 \times b = 0 = -(a \times b) + a \times b$$

3. If R has identity then $(-1) \times a = -a$ for $a \in R$

4. 加法反元素 $\neq$ 乘法反元素 ($1 \neq 0$)

1.3.3. Subring

A subset S of a ring R is a **subring** if S itself is a ring with $+, \times$ in R

Subring 的 Example

- $\mathbb{Z}$ is a subring of $\mathbb{Q}$ or $\mathbb{R}$
- $\mathbb{Q}[i]$ is a subring of $\mathbb{C}$ ($\mathbb{Q}[i]$ 是 $a + bi$, a 和 b 都是有理數)
- $\mathbb{C}[x]$ is a subring of $\mathbb{C}[x, y]$
- $6\mathbb{Z}$ is a subring of $2\mathbb{Z}$

!! 從這邊之後的 ring 都是指 commutative ring with identity !!

1.3.4. Zero Divisor

If non-zero $a, b \in R$ satisfy $a \times b = 0$, then a and b are zero divisors of R

Zero Divisor 的 Example

- 2, 3, 4 are zero divisors in $(\mathbb{Z}_6, +, \times)$ ($2 \times 3 \bmod 6 = 0, 3 \times 4 \bmod 6 = 0$)
- $\mathbb{Z}_p$ has no zero divisors for every prime p
- $\mathbb{Z}_n$ with usual operations has zero divisors iff n is composite (合數)

1.3.5. Integral Domain

A ring is an **integral domain** if R has no zero divisors

Integral Domain 的 Example

- $\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}, \mathbb{Q}[x], \mathbb{Z}[x, y]$ are integral domains
- $\mathbb{Z}[\sqrt{2}] = \{a + b\sqrt{2} \mid a, b \in \mathbb{Z}\}$ is an integral domain
- $\mathbb{Z}_6$ is not an integral domain

The Number Field Sieve is based on the theory with Dedekind Domains, Unique Factorization Domains (UFD), and Principal Ideal Domains (PID)

1.3.6. Ideal

A subset I is an **ideal** of a ring (commutative, 有 identity 的) R if :

1. I is an additive group
 2. $ar \in I$ for all $a \in I$ and $r \in R$
- Every ring R has at least two ideals, $\{0\}$ and R
 - $aR := \{ar \mid r \in R\}$ is an ideal of R for every $a \in R$
 - The ideal $\langle a \rangle := \{ar \mid r \in R\}$ of a ring R is a **principal ideal** generated by $a \in R$

Ideal 的 Example

- $n\mathbb{Z}$ is an ideal of $\mathbb{Z}$ for every $n \in \mathbb{Z}$
- $I = \{h(x)(x^2 - 2) \mid h(x) \in \mathbb{Q}[x]\}$ is an ideal of $\mathbb{Q}[x]$
- $I = \text{"the set of polynomials in } \mathbb{Z}[x] \text{ with constant terms being even integers"}$ is an ideal of $\mathbb{Z}[x]$
- Let I_1 and I_2 be two ideals of R , then :
 - $I_1 \cap I_2$ is an ideal of R
 - $I_1 + I_2 := \{a_1 + a_2 \mid a_1 \in I_1, a_2 \in I_2\}$ is an ideal of R

Let I_1 and I_2 be two ideals of R , then :

- $I_1 \cap I_2$ is an ideal of R
- $I_1 + I_2 := \{a_1 + a_2 \mid a_1 \in I_1, a_2 \in I_2\}$ is an ideal of R

e.g. $R = \mathbb{Z}, I_1 = 4\mathbb{Z}, I_2 = 3\mathbb{Z}, I_3 = 6\mathbb{Z}$

- $I_1 \cap I_2 = 12\mathbb{Z}, I_1 \cap I_3 = 12\mathbb{Z}$
- $I_1 + I_2 = \mathbb{Z}, I_1 + I_3 = 2\mathbb{Z}$

不是 Ideal 的 Example

- $\mathbb{Z}$ is not an ideal of $\mathbb{Q}$
 - $2 \in \mathbb{Z}, \frac{1}{3} \in \mathbb{Q}$, but $2 \times \frac{1}{3} \notin \mathbb{Z}$
- $\mathbb{Q}$ is not an ideal of $\mathbb{R}$
- $\mathbb{C}[x]$ is a subring, but not an ideal of $\mathbb{C}[x, y]$

Principal Ideal 的 Example

- $\langle 1 \rangle = R$ is a type of principal ideal of R
- $\langle x \rangle$ and $\langle x^2 + 1 \rangle$ are principal ideals of $\mathbb{Z}[x]$
- every ideal in $\mathbb{Z}$ is principal ideal
 - $\langle 15, 21 \rangle (= \{15m + 21n \mid m, n \in \mathbb{Z}\}) = \langle 3 \rangle$ is principal
 - $\mathbb{Z}$ is a PID (Principal Ideal Domain)
- $\langle x, y \rangle$ in $\mathbb{Q}[x, y]$ is **not** a principal ideal
- the ideal consisting of polynomials in $\mathbb{Z}[x]$ with even constant terms is **not** a principal ideal (proof 在下面)

Proof

suppose the ideal consisting of polynomials in $\mathbb{Z}[x]$ with even constant terms is $\langle a \rangle$
 $\Rightarrow 2 = af(x)$ and $x = ag(x)$ for some $f(x), g(x) \in \mathbb{Z}[x]$ ($x = x + 0$ also even constant)
 $\Rightarrow \deg(2) = \deg(a) + \deg(f(x)) \Rightarrow 0 = \deg(a) + \deg(f(x)) \Rightarrow \deg(a), \deg(f(x)) = 0$
 $\Rightarrow a$ is a constant $\rightarrow a = 0 \vee 2$ (因為 $af(x) = 2$)

1. $a = 1 \Rightarrow \langle a \rangle = \mathbb{Z}[x] \Rightarrow$ 矛盾
2. $a = 2 \Rightarrow \langle a \rangle = 2\mathbb{Z}[x] \Rightarrow$ 和 $x = g(x)$ 矛盾

1.3.7. Congruent Class and Quotient Ring

- If I is an ideal of a ring R and $a, b \in R$, then a is **congruent to $b \pmod I$** if $a - b \in I$
- If I is an ideal of a ring R and $a \in R$, then the **congruence class** of $a \pmod I$, denoted as $a + I$, is the set of all elements of R congruent to $a \pmod I$
 - The congruence class $a + I$ is a **coset** of I in R

$$R = \mathbb{Z}, I = \langle 4 \rangle = 4\mathbb{Z} = \{4r \mid r \in \mathbb{Z}\}$$

- example of cosets :
 - $2 + I = \{\dots, -6, -2, 2, 6, 10, \dots\}$
 - $3 + I = \{\dots, -5, -1, 3, 7, 11, \dots\}$
- it's simple to define $+$ and $\times$ on them :
 - $(2 + I) + (3 + I) = (1 + I)$ since $2 + 3 \pmod 4 = 1$
 - $(2 + I) \times (3 + I) = (2 + I)$ since $2 \times 3 \pmod 4 = 2$

If I is an ideal of a ring R , then the set of all cosets of I form a ring called the **quotient ring** of R by I , denoted as $\frac{R}{I}$, with the operations :

1. $(a + I) + (b + I) = (a + b) + I$
2. $(a + I) \times (b + I) = (a \times b) + I$

Quotient Ring 的 Example

- R is a ring, then $\frac{R}{R} \cong \{0\}$ and $\frac{R}{\{0\}} = R$
- $R = \mathbb{R}[x], I = \langle x^2 + 1 \rangle$, then $\frac{R}{I} \cong \mathbb{C}$
- $\frac{\mathbb{Z}}{n\mathbb{Z}} \cong \mathbb{Z}_n$

1.3.8. Homomorphism and Isomorphism

$(R_1, +_1, \times_1)$ and $(R_2, +_2, \times_2)$ are two rings

- $f : R_1 \rightarrow R_2$ is a homomorphism if :
 1. $f(a +_1 b) = f(a) +_2 f(b)$ for all $a, b \in R$
 2. $f(a \times_1 b) = f(a) \times_2 f(b)$ for all $a, b \in R$
- the ideal $\ker(f) =: \{r \in R : f(r) = 0\}$ is the kernel of f
- $f : R_1 \rightarrow R_2$ is a isomorphism (R is isomorphic to S , or $R \cong S$) if :
 1. f is homomorphism
 2. f is bijective : injective (one-to-one) and surjective (onto)

Homomorphism and Isomorphism 的 Example

- complex conjugate map $f : \mathbb{C} \rightarrow \mathbb{C}$ defined by $f(a + bi) = a - bi$ is an isomorphism
 - it is also an automorphism (an isomorphism to itself)
- the zero map $f : \mathbb{Z} \rightarrow \mathbb{Z}$ defined by $f(a) = 0$ for all $a \in R$ is a homomorphism
 - but not isomorphism
- $f : \mathbb{Z} \rightarrow \mathbb{Z}_n$ defined by $f(x) = x \bmod n$ for some fixed n is a homomorphism
 - but not isomorphism (surjective but not injective)
- $f : \mathbb{Q}[\omega] \rightarrow \mathbb{C}$ defined by $f(x) = x$ is a homomorphism
 - but not isomorphism (injective but not surjective)
 - $\mathbb{Q}[\omega] = \{a + b\omega \mid a, b \in \mathbb{Q}\}$ where ω is a root of $x^2 + x + 1 = 0$

1.3.9. Chinese Remainder Theorem

Proposition

Let $I_1, I_2, I_3, \dots, I_k$ be ideals of R

the map $R \rightarrow \frac{R}{I_1} \times \frac{R}{I_2} \times \dots \times \frac{R}{I_k}$ defined by $r \mapsto (r + I_1, r + I_2, \dots, r + I_k)$ is a ring homomorphism with kernel $I_1 \cap I_2 \cap \dots \cap I_k$

If for each $i, j \in \{1, 2, \dots, k\}$ with $i \neq j$, the ideal $I_i + I_j = R$, then this map is surjective that is, $\frac{R}{I_1 \cap I_2 \cap \dots \cap I_k} \cong \frac{R}{I_1} \times \frac{R}{I_2} \times \dots \times \frac{R}{I_k}$

e.g. $\frac{\mathbb{Z}}{35\mathbb{Z}} \cong \frac{\mathbb{Z}}{5\mathbb{Z}} \times \frac{\mathbb{Z}}{7\mathbb{Z}}$

1.4. Field

1.4.1. Definition

A ring R is a **field** if every nonzero $a \in R$ has an inverse a^{-1}

Field 的 Example

- $\mathbb{Q}, \mathbb{R}, \mathbb{C}$ are fields
- $\mathbb{Z}_p$ with prime p is a field
 - also denoted as $\mathbb{F}_p$ or $GF(p)$, Galois field of order p
- $GF(2^8)$ is used in AES
- $\mathbb{Z}$ is not a field ($2^{-1} \notin \mathbb{Z}$)

1.4.2. Characteristic of a Field

The **characteristic** of a field F , denoted as $ch(F)$, is defined to be the smallest positive integer p s.t. $p \cdot 1_F = 0$ if such a p exists, and is defined to be 0 otherwise

Proposition

Let F be a field with characteristic $p > 0$, then :

- p is a prime

Proof

suppose not, let $p = q_1 q_2, q_1, q_2 \geq 2$

$$0 = p \cdot 1_F = (q_1 q_2) \cdot 1_F = (q_1 \cdot 1_F)(q_2 \cdot 1_F)$$

$\therefore q_1 \cdot 1_F = 0$ or $q_2 \cdot 1_F = 0$, contradict to the minimality of p

- $p \mid \binom{p}{i}$ for $2 \leq i \leq p-1$
- $(a+b)^p = \sum_{i=0}^p \binom{p}{i} a^i b^{p-i} = a^p + b^p$
- $(a+b)^{p^2} = [(a+b)^p]^p = (a^p)^p + (b^p)^p = a^{p^2} + b^{p^2}$
- $(a^p)^{p^n} = a^{p^{p^n}} + b^{p^{p^n}}$ by induction

1.4.3. Subfield

A subset K of a field F is a **subfield** of F if K is itself a field with $+, \times$ in F

- also F is called the **field extension** of K

Subfield 的 Example

- $\mathbb{Q}$ is a subfield of $\mathbb{R} \Leftrightarrow \mathbb{R}$ is a field extension of $\mathbb{Q}$
- $\mathbb{R}$ is a subfield of $\mathbb{C} \Leftrightarrow \mathbb{C}$ is a field extension of $\mathbb{R}$
 - $\mathbb{C} = \{a + bi \mid a, b \in \mathbb{R}\}$ can be considered as a vector space over $\mathbb{R}$ of dimension 2 with basis $\{1, i\}$, hence $\mathbb{C}$ is a field extension of $\mathbb{R}$ with degree 2

1.4.4. Irreducible Polynomial

A non-constant polynomial $p(x) \in F[x]$ (F is a field) is **irreducible** if its only divisors are its associates ($cp(x)$) and non-zero constants

Irreducible 的 Example

- $x^2 + 2$ is irreducible in $\mathbb{Z}_5[x]$
- $x^2 + 2$ is reducible in $\mathbb{Z}_3[x]$
 - since $x^2 + 2 = (x+1)(x+2)$
- $x^2 + 1$ is irreducible in $\mathbb{Z}[x]$, but is reducible in $\mathbb{C}[x]$

1.4.5. Field Extension

A **field extension** is a pair of fields (K, L) where K is a subfield of L (denoted as $L : K$)

- 通常 L 大 K 小, 值就是 $\frac{L}{K}$
- If $L : K$ is a field extension, then L has a natural structure as a vector space over K
- The dimension of this vector space is called the **degree** of the extension, or the degree of L over K , which is written as $[L : K]$

Proposition

- If (K, L) and (L, M) are field extensions, (K, M) is also a one, $[M : K] = [M : L][L : K]$

$$\text{e.g. } [\mathbb{Q}(\sqrt{2}, \sqrt{3}) : \mathbb{Q}] = [\mathbb{Q}(\sqrt{2}, \sqrt{3}) : \mathbb{Q}(\sqrt{2})] [\mathbb{Q}(\sqrt{2}) : \mathbb{Q}] = 2 \times 2 = 4$$

- Let $p(x) \in F[x]$ be an irreducible polynomial, then $\frac{F[x]}{\langle p(x) \rangle}$ is a field extension of F

Remark

- $\frac{\mathbb{R}[x]}{\langle x+1 \rangle} \cong \mathbb{R} \cong \frac{\mathbb{R}}{\langle x-1 \rangle}$
- $\frac{\mathbb{R}[x]}{x^2+1} \cong \mathbb{C}$
- by Chinese Remainder Theorem, $\frac{\mathbb{R}[x]}{\langle x^2-1 \rangle} \cong \frac{\mathbb{R}[x]}{\langle x+1 \rangle} \times \frac{\mathbb{R}[x]}{\langle x-1 \rangle} \cong \mathbb{R} \times \mathbb{R}$
- $\mathbb{C} \neq \mathbb{R} \times \mathbb{R}$ since $\mathbb{R} \times \mathbb{R}$ has zero divisors ($\mathbb{R} \times \mathbb{R}$ 是 $a \times b$ 這種形式)

1.4.6. Simple Extension

If the field L is generated by a single element a over K , $L = K(a)$, then L is said to be a **simple extension**

Proposition

Let (K, L) be a field extension,

then for any $a \in L$, $K(a) = \left\{ \frac{f(a)}{g(a)} \mid f(x), g(x) \in \mathbb{K}[x], g(x) \neq 0 \right\}$ is a field

$$\text{e.g. } \mathbb{Q}(\sqrt{2}) = \left\{ \frac{a+b\sqrt{2}}{c+d\sqrt{2}} \mid a, b, c, d \in \mathbb{Q}, c+d\sqrt{2} \neq 0 \right\} \text{ is a field}$$

1.4.7. Primitive Element Theorem

Let (K, L) be a field extension of finite degree and $\text{ch}(K) = 0$, then $L = K(a)$ for some $a \in L$
i.e. any finite extension of fields of characteristic 0 is simple

2. Arithmetic

2.1. Modular Function

Floor and Ceiling

- the **floor** $\lfloor x \rfloor$ of $x \in \mathbb{R}$ is the largest integer $\leq x$
- the **ceiling** $\lceil x \rceil$ is the smallest integer $\geq x$

e.g. $\lfloor e \rfloor = 2, \lceil e \rceil = 3, \lfloor -3.14 \rfloor = -4, \lceil -3.14 \rceil = 3$

e.g. $25 = 3_{(\text{quotient})} \times 7_{(\text{divisor})} + 4_{(\text{remainder})} \Rightarrow 25 \bmod 7 = 25 - \lfloor \frac{25}{7} \rfloor \times 7$

Modular Function

$m, n \in \mathbb{Z}, m > 0$, define $n \bmod m = n - \lfloor \frac{n}{m} \rfloor \times m$ (也就是 $\frac{n}{m}$ 之後的餘數)

e.g. 58 in the base 3 representation :

$$\lfloor \frac{58}{3} \rfloor = 19, 58 \bmod 3 = 1$$

$$\lfloor \frac{19}{3} \rfloor = 6, 19 \bmod 3 = 1$$

$$\lfloor \frac{6}{3} \rfloor = 2, 6 \bmod 3 = 0$$

$$\lfloor \frac{2}{3} \rfloor = 0, 2 \bmod 3 = 2$$

$$\Rightarrow (58)_{10} = (2011)_3$$

Cayley Tables

Define $\oplus$ and $\otimes$ on $\mathbb{Z}_6 = \{0, 1, 2, 3, 4, 5\}$ as :

$$a \oplus b = (a + b) \bmod 6$$

$\oplus$	0	1	2	3	4	5
0	0	1	2	3	4	5
1	1	2	3	4	5	0
2	2	3	4	5	0	1
3	3	4	5	0	1	2
4	4	5	0	1	2	3
5	5	0	1	2	3	4

$$a \otimes b = (a \times b) \bmod 6$$

$\otimes$	0	1	2	3	4	5
0	0	0	0	0	0	0
1	0	1	2	3	4	5
2	0	2	4	0	2	4
3	0	3	0	3	0	3
4	0	4	2	0	4	2
5	0	5	4	3	2	1

$\Rightarrow \mathbb{Z}_6$ 不是 field 因為 2 沒有乘法反元素

2.2. GCD and Euclidean Algorithm

2.2.1. Greatest Common divisor (GCD)

Assume $a, b \in \mathbb{Z}, a \neq 0$ or $b \neq 0$:

- $d \neq 0$ is a **common divisor** of a and b if $d \mid a$ and $d \mid b$
- The **greatest common divisor** $d = \gcd(a, b)$ is the largest of the common divisors

e.g. divisors of 20 : $\pm 1, \pm 2, \pm 5, \pm 10, \pm 20$

2.2.2. Proposition

if $a \in \mathbb{P}$

- $\gcd(a, a) = a$
- $\gcd(a, 0) = a$

if $a, b \in \mathbb{Z}, a \neq 0$ or $b \neq 0$

- $\gcd(a, b) = \gcd(|a|, |b|)$
- $\gcd(a, b) = \gcd(b, a)$

$a, b \in \mathbb{Z}, a \neq 0$ or $b \neq 0$, then $\gcd(a, b) = \gcd(a + kb, b)$ for any $k \in \mathbb{Z}$ (這是 Thm 不是 Prop)

Proof

Define $A = a$ 和 b 的公因數的 set, $B = a + kb$ 和 b 的公因數的 set

(i) $A \subset B$: 假設 $d \in A \Rightarrow d \mid a, d \mid b \Rightarrow a = xd, b = yd$ for some $x, y \in \mathbb{Z}$

$$\Rightarrow a + kb = xd + k(yd) = (x + ky)d \Rightarrow d \mid (a + kb) \Rightarrow d \in B$$

(ii) $B \subset A$: $\Rightarrow c \in B$, 這樣 $c \mid (a + kb), c \mid b \Rightarrow a + kb = xc, b = yc$ for some $x, y \in \mathbb{Z}$

$$\Rightarrow a = xc - kb = xc - k(yc) = (x - ky)c \Rightarrow c \mid a \Rightarrow c \in A$$

$$\Rightarrow A = B, \gcd(a, b) = \gcd(a + kb, b)$$

Corollary : 如果 $b > 0$, then $\gcd(a, b) = \gcd(b, a \bmod b)$

Proof

$$\gcd(a, b) = \gcd(a - \lfloor \frac{a}{b} \rfloor \times b, b)_{\text{Thm}} = \gcd(a \bmod b, b)_{\text{Def}} = \gcd(b, a \bmod b)_{\text{Prop}}$$

2.2.3. Euclidean Algorithm

歐幾里德演算法 aka 輾轉相除法

直接算

$$\begin{aligned} &\gcd(325, 234) \\ &= \gcd(234, 325 \bmod 234) \\ &= \gcd(234, 91) \\ &= \gcd(91, 234 \bmod 91) \\ &= \gcd(91, 52) \\ &= \gcd(52, 91 \bmod 52) \\ &= \gcd(52, 39) = \dots \end{aligned}$$

直式

1	325	234	2
	234	182	
1	91	52	1
	52	39	
3	39	13	
	39		
	0		

Pseudo Code

normal way

```

1 if b = 0
2   gcd = |a|
3 else
4   while b != 0
5     c = b
6     b = a mod b
7     a = c
8   return gcd

```

recursive way

```

1 if b = 0
2   gcd = |a|
3 else
4   gcd = gcd(b, a % b)
5   return gcd

```

2.3. Extended GCD Algorithm

Algorithm : Given $a, b \in \mathbb{Z}$, find $x, y \in \mathbb{Z}$ s.t. $ax + by = \gcd(a, b)$

Application : b 是質數 $\Rightarrow \gcd(a, b) = 1 \Rightarrow a, b$ 互質

法一 $\Rightarrow a \bmod b$ 乘法反元素存在 (假設叫 x) $\Rightarrow ax \equiv 1 \pmod{b}$

法二 $\Rightarrow ax + by = 1 \Rightarrow ax + by \pmod{b} = 1 \pmod{b} \Rightarrow ax \equiv 1 \pmod{b}$

e.g. $a = 100, b = 35$

$$100 = 2 \times 35 + 30$$

$$35 = 1 \times 30 + 5$$

$$30 = 6 \times 5 + 0$$

$$\Rightarrow \gcd(100, 35) = 5 = 35 - 30 = 35 - (100 - 2 \times 35) = (-1) \times 100 + (3) \times 35$$

$$\Rightarrow x = -1, y = 3$$

先用輾轉相除法找 **gcd** 再原路代回去

Theorem : given $a, b \in \mathbb{Z}$, there $\exists x, y \in \mathbb{Z}$ for the **linear combination** $ax + by = \gcd(a, b)$

Proof

$$\text{take } r_0 = a = a \times 1_{[=x_0]} + b \times 0_{[=y_0]} = ax_0 + by_0$$

$$r_1 = b = a \times 0_{[=x_1]} + b \times 1_{[=y_1]} = ax_1 + by_1$$

for $i > 0$, let $q_{i+1} = \lfloor \frac{r_{i-1}}{r_i} \rfloor$

$$r_{i+1} = r_{i-1} - r_i q_{i+1}$$

$$x_{i+1} = x_{i-1} - x_i q_{i+1}$$

$$y_{i+1} = y_{i-1} - y_i q_{i+1}$$

assume $r_{i-1} = ax_{i-1} + by_{i-1}$ and $r_i = ax_i + by_i$

$$\text{then } r_{i+1} = r_{i-1} - r_i q_{i+1} = (ax_{i-1} + by_{i-1}) - (ax_i + by_i)q_{i+1} = ax_{i+1} + by_{i+1}$$

$\Rightarrow$ 由數學歸納法得證

Algorithm (matrix version)

$$1 \text{ set } \begin{bmatrix} r_0 & x_0 & y_0 \\ r_1 & x_1 & y_1 \end{bmatrix} = \begin{bmatrix} a & 1 & 0 \\ b & 0 & 1 \end{bmatrix}$$

2 while $r \neq 0$

$$3 \text{ set } \begin{bmatrix} r_0 & x_0 & y_0 \\ r_1 & x_1 & y_1 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & -\lfloor \frac{r_0}{r_1} \rfloor \end{bmatrix} \begin{bmatrix} r_0 & x_0 & y_0 \\ r_1 & x_1 & y_1 \end{bmatrix}$$

$$4 \text{ return } [d, x, y] = [r_0, x_0, y_0]$$

2.4. Stein's GCD Algorithm / Stein's Recursion

three least-significant-bit GCD properties

- a, b both even $\Rightarrow \gcd(a, b) = 2 \gcd(\frac{a}{2}, \frac{b}{2})$
- a is even, b is odd $\Rightarrow \gcd(a, b) = \gcd(\frac{a}{2}, b)$
- a, b both odd $\Rightarrow \gcd(a, b) = \gcd(\frac{a-b}{2}, b)$

一直遞迴下去直到找到 $\gcd$ 為止

因為除以 2 只是往右 shift 一格而已, 沒有一般的除法麻煩 $\Rightarrow$ 目前最快的 gcd algorithm

2.5. Many Theorems of Arithmetic

- **Theorem** : $\gcd(a, b)$ is the smallest positive integer that can be written as $d = ax + by$ where $x, y \in \mathbb{Z}$
- **Definition** : a and b are **relatively prime** ($a \perp b$) if $\gcd(a, b) = 1$
- **Theorem** : If $a \mid (bc)$ and $a \perp b$ then $a \mid c$

Proof

$a \perp b \Rightarrow ax + by = 1$ for some $x, y \in \mathbb{Z}$

multiply by $c \Rightarrow acx + bcy = c \Rightarrow a \mid (bc) \Rightarrow a \mid (bcy) \Rightarrow a \mid (acx + bcy) = c$

- **Corollary** : p is a prime and $p \mid (ab)$, then $p \mid a$ or $p \mid b$

Proof

兩種可能 $\Rightarrow$ 1. $p \mid a \Rightarrow$ done, 2. $p \nmid a \Rightarrow \gcd(p, a) = 1 \Rightarrow p \mid b$ by theorem

- **Corollary** : p is a prime and $p \mid (a_1 a_2 a_3 \dots a_k)$, then $p \mid a_i$ for some i

Proof

一直做上一個 corollary 的動作就好

- **Theorem (The Fundamental Theorem of Arithmetic)** : Every integer $a > 1$ is itself prime or can be written unique as a product of primes

Proof

Existence : 一直做上一個 corollary 的動作就好

Uniqueness : 用上一個 corollary

- **Definition** : Unique Factorization Domain (UFD) 代表有唯一質因數分解的集合

UFD 的 Example

- ▶ $\mathbb{Z}$ is a UFD
- ▶ $\mathbb{Z}[\sqrt{-5}] = \{a + b\sqrt{-5} \mid a, b \in \mathbb{Z}\}$ is not UFD, 例如 6 就有兩種分解法
 $\Rightarrow 6 = 2 \times 3 = (1 - \sqrt{-5})(1 + \sqrt{-5}) \Rightarrow$ 都已經沒辦法分解了 (irreducible)

- **Definition** : The least common multiple $\text{lcm}(a, b)$ of $a, b \in \mathbb{P}$ is the smallest $m \in \mathbb{P}$ s.t. $a \mid m$ and $b \mid m$
- **Theorem** : $a, b \in \mathbb{P}$, then $a \times b = \gcd(a, b) = \text{lcm}(a, b)$

2.6. Modular Arithmetic

2.6.1. Congruence Relation (同餘)

Given $a, b \in \mathbb{Z}, m \in \mathbb{P}$, then a is congruent to b modulo m ($a \equiv b(\text{mod } m)$)

- $a \equiv b$ iff $m \mid (a - b)$

2.6.2. Proposition

- if $a, b, c \in \mathbb{Z}, m, d \in \mathbb{P}$, then
 - $a \equiv a(\text{mod } m)$
- if $a \equiv b(\text{mod } m)$, then
 - $b \equiv a(\text{mod } m)$
 - $a + c \equiv b + c(\text{mod } m)$
 - $ac \equiv bc(\text{mod } m)$
 - $a^d \equiv b^d(\text{mod } m)$

Proof

$$a \equiv b(\text{mod } m) \Leftrightarrow m \mid (a - b)$$

$$\Rightarrow m \mid (a - b)(a^{d-1} + a^{d-2}b + \dots + b^{d-1}) = a^d - b^d$$

- if $a \equiv b$ and $b \equiv c(\text{mod } m)$, then $a \equiv c(\text{mod } m)$
- if $d \equiv \text{gcd}(a, m)$, $ab \equiv ac(\text{mod } m)$, then $b \equiv c(\text{mod } \frac{m}{d})$

Proof

we have $d = ax + my$ for some $x, y \in \mathbb{Z}$

Hence $(\frac{a}{d})x + (\frac{m}{d})y = 1$ where $\frac{a}{d}, \frac{m}{d} \in \mathbb{Z}$

multiply by $(b - c) \Rightarrow (\frac{a}{d})(b - c)x + (\frac{m}{d})(b - c)y = b - c$

by assumption $a \mid a(b - c) \Rightarrow (\frac{m}{d}) \mid (\frac{a}{d})(b - c)$

$\Rightarrow (\frac{m}{d}) \mid [(\frac{a}{d})(b - c)x + (\frac{m}{d})(b - c)y] = b - c$

2.6.3. Inverse

The inverse of a modulo m ($a^{-1} \text{ mod } m$) is the smallest positive integer s.t. $ac \equiv 1(\text{mod } m)$

- $a^{-1} \text{ mod } m$ exists iff $a \perp m$
- the solution to linear equation $ax \equiv b(\text{mod } m)$ is $x \equiv a^{-1}(ax) \equiv a^b(\text{mod } m)$

Algorithm (matrix version)

```

1 set  $\begin{bmatrix} r_0 & x_0 \\ r_1 & x_1 \end{bmatrix} = \begin{bmatrix} a & 1 \\ m & 0 \end{bmatrix}$ 
2 while  $r_1 \neq 0$ 
3   set  $\begin{bmatrix} r_0 & x_0 \\ r_1 & x_1 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & -\lfloor \frac{r_0}{r_1} \rfloor \end{bmatrix} \begin{bmatrix} r_0 & x_0 \\ r_1 & x_1 \end{bmatrix}$ 
4 if  $r_0 = 1$ 
5   return  $x_0 \equiv a^{-1} \text{ mod } m$ 
6 else return "no inverse"
```

Example - solve $160x \equiv 7(\text{MOD } 841)$

$$\begin{bmatrix} r_0 & a_0 \\ r_1 & x_1 \end{bmatrix} = \begin{bmatrix} a & 1 \\ m & 0 \end{bmatrix} = \begin{bmatrix} 160 & 1 \\ 841 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 \\ 1 & -\lfloor \frac{160}{841} \rfloor \end{bmatrix} \begin{bmatrix} 160 & 1 \\ 841 & 0 \end{bmatrix} = \begin{bmatrix} 841 & 0 \\ 160 & 1 \end{bmatrix}$$

...

$$\begin{bmatrix} 0 & 1 \\ 1 & -\lfloor \frac{4}{1} \rfloor \end{bmatrix} \begin{bmatrix} 4 & -21 \\ 1 & 205 \end{bmatrix} = \begin{bmatrix} 1 & 205 \\ 0 & -841 \end{bmatrix}$$

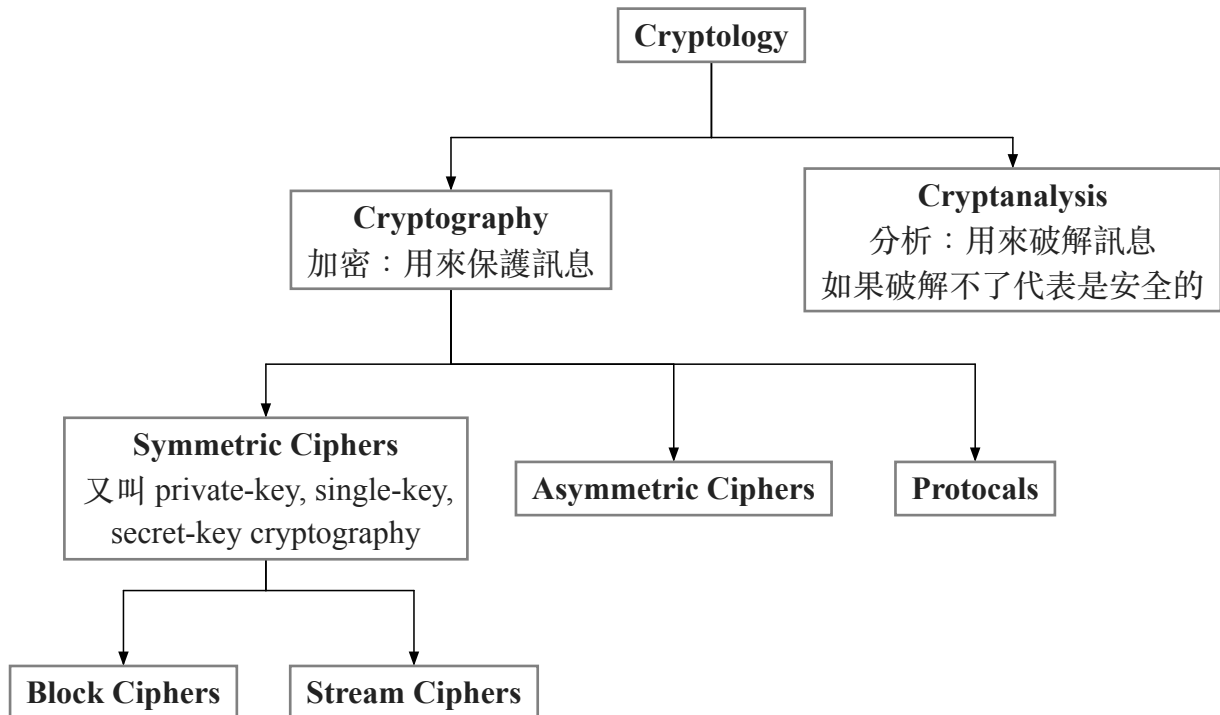
$$\Rightarrow 160^{-1} \equiv 205(\text{mod } 841)$$

$$\Rightarrow x \equiv 160^{-1} \times 7 \equiv 205 \times 7 \equiv 594(\text{mod } 841)$$

3. Finite Fields

4. Cryptography

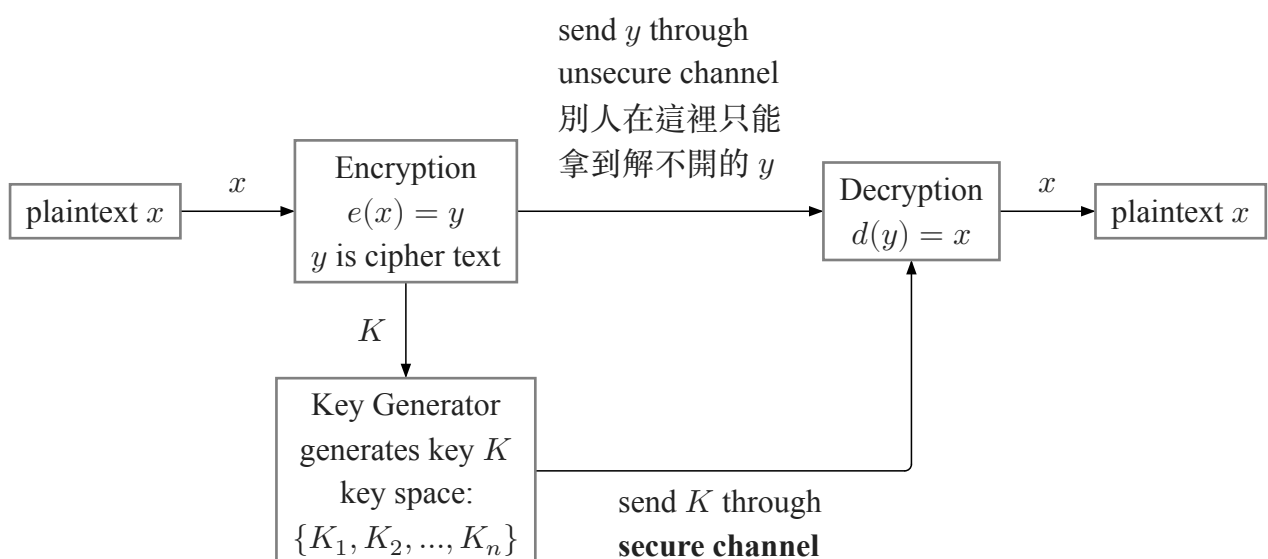
4.1. Introduction



Basic Facts

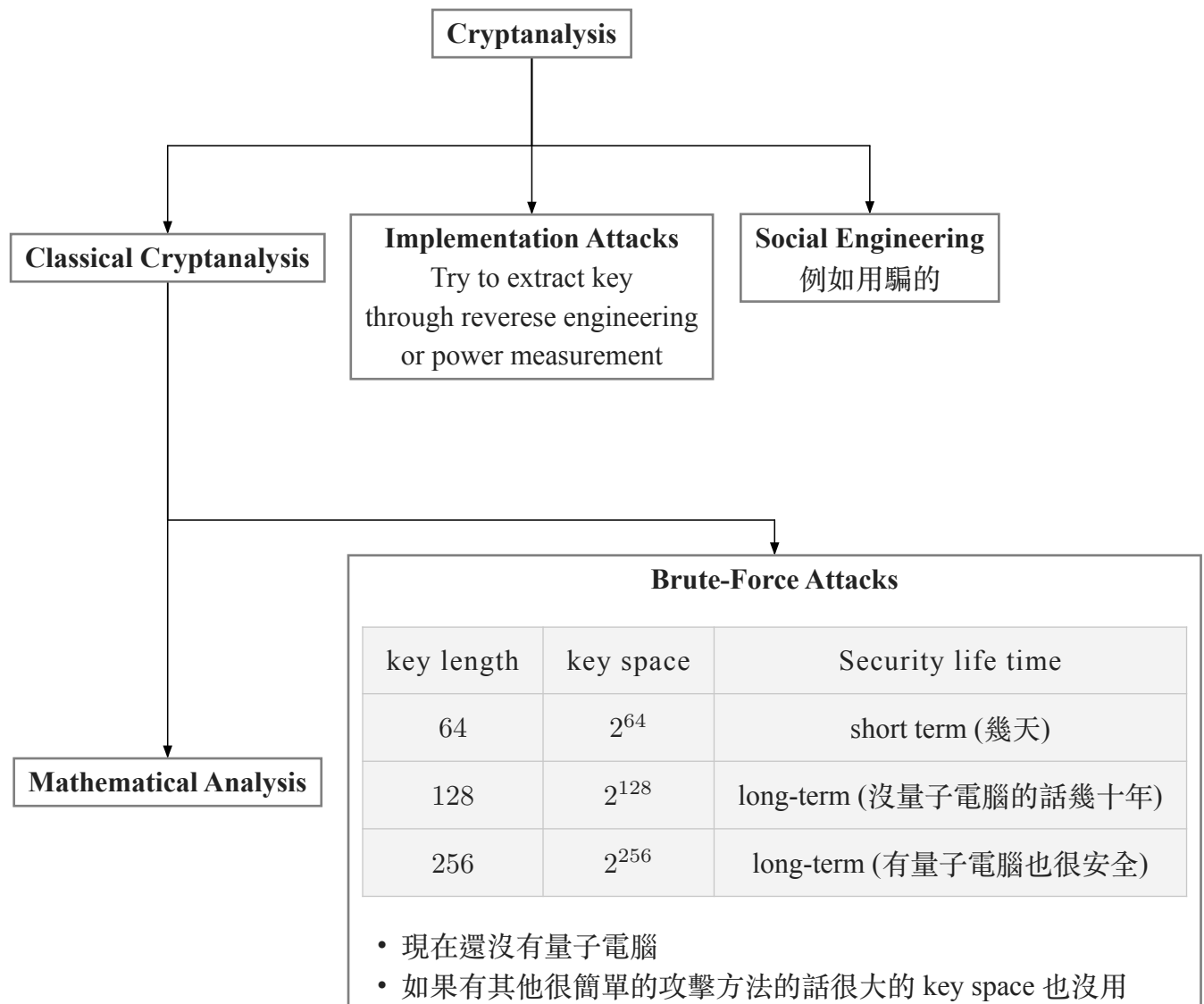
- Ancient Crypto: 2000 B.C. 古埃及 Letter-based encryption scheme 就很紅 (Ceasar cipher)
- Symmetric Ciphers: 從古時候到 1976 的加密都是 Symmetric 的
- Asymmetric Ciphers: 1976 Diffie, Hellman, Merkle 發明了 public-key cryptography
- Hybrid Schemes: 現在用的 protocol 大部分是都有用的
 - symmetric ciphers for encryption and message authentication
 - asymmetric ciphers for key exchange and digital signature

4.1.1. Symmetric Cryptography



encryption equation and decryption equation are inverse operations: $d_K(y) = d_K(e_K(x)) = x$

4.1.2. Cryptanalysis



4.1.3. Substitution Cipher

方法：把每個字母都選一個特定的字母換掉
有兩種解法：

Exhaustive Key Search (Brute-Force)	Letter Frequency Analysis (Brute-Force)
一個一個找 有 $26 \times 25 \times \dots \times 3 \times 2 \times 1 = 2^{88}$ keys substitution table (=keys) 超大	出現頻率最高的英文字母是 e (大概 13%) 方法： 1. 用 e 換掉在 cipher text 裡面出現頻率最高的字母 2. 再用頻率次高的字母換掉 cipher text 裡面出現第二多次的字母 ⇒ 要試的次數少很多

4.1.4. Shift (Ceasar) Cipher and Affine Cipher

兩種方法都會用把字母對應到數字 $\Rightarrow A = 0, B = 1, \dots, Z = 25$

Shift (Ceasar) Cipher

把每個字母用它 $+ \text{key} \bmod 26$ 換掉

Let $k, x, y \in \{0, 1, \dots, 25\}$

- Encryption: $y = e_k(x) \equiv x + k \bmod 26$
- Decryption: $x = d_k(y) \equiv y - k \bmod 26$

key space is only 26

Affine Cipher

比 Shift Cipher 更高級一點

key : (a, b)

Let $a, b, x, y \in \{0, 1, \dots, 25\}$

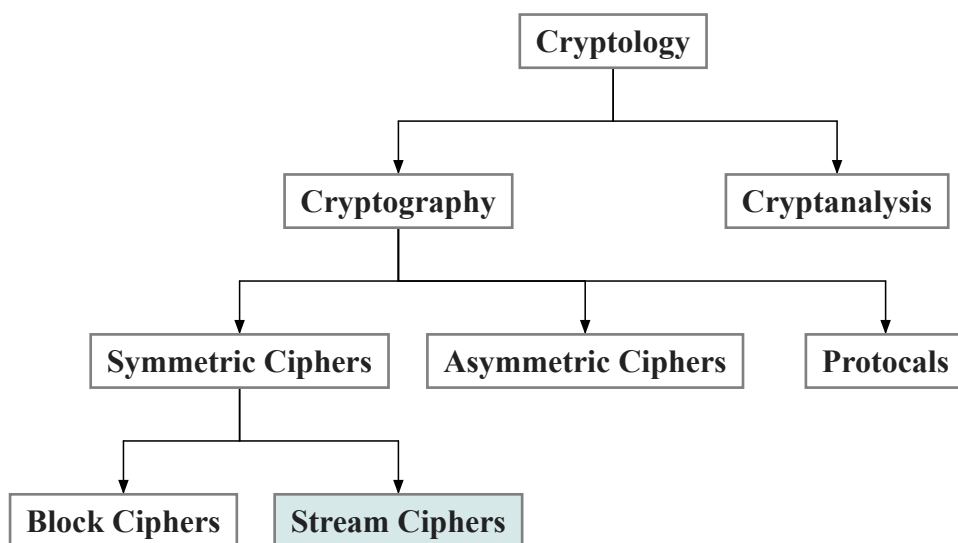
- Encryption: $y = e_k(x) \equiv ax + b \bmod 26$
- Decryption: $x = d_k(y) \equiv a^{-1}(y - b) \bmod 26$

a 只能選 $\gcd(a, 26) = 1$ 的 不然沒有 inverse (總共有 12 個)

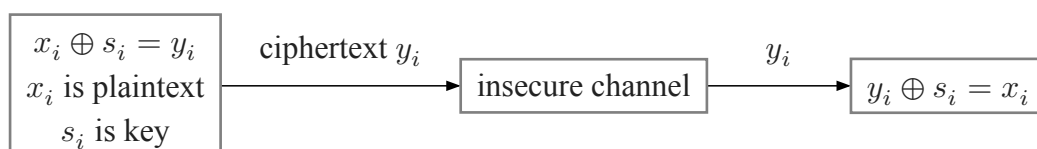
key space is $12 \times 26 = 312$

4.2. Stream Ciphers

4.2.1. Introduction to Stream Ciphers



Stream Ciphers	Block Ciphers
一次只加密一個字元 small and fast $\Rightarrow$ common in embedded devices (e.g. A5/1 for GSM phones)	一次加密一整個 block are common for Internet applications



特性：

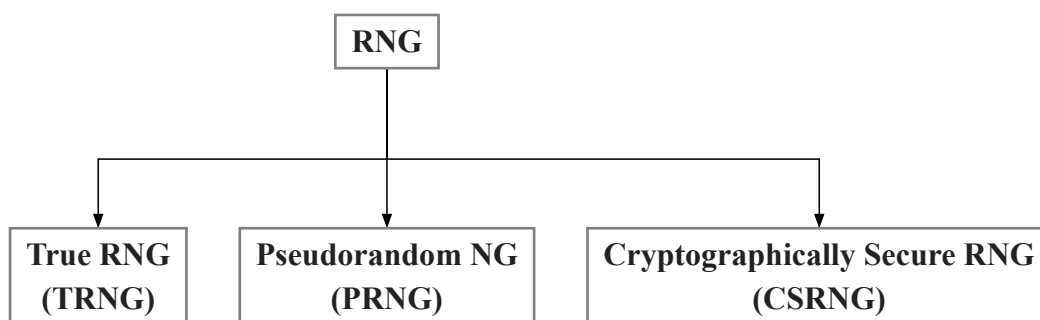
- XOR 的運算 ($\oplus$) 跟加起來 mod 2 是一樣的 $\Rightarrow$ 很簡單
- encryption and decryption are the same functions / inverting XOR is same as XOR itself ($x_i, y_i, s_i \in \{0, 1\}$):
 - encryption : $y_i = e_{s_i}(x_i) = x_i + s_i \bmod 2$
 - decryption : $x_i = d_{s_i}(y_i) = y_i + s_i \bmod 2$
- output bit 是 0 或 1 的機率都是 50% $\Rightarrow$ good statistic property

Synchronous Stream Cipher	Asynchronous Stream Cipher
Key stream depends only on the key (and possibly initialization vector)	Key stream depends also on the ciphertext 也就是一個位元形成密文之後會丟去 key generator 來生成後面的 key 所以後面的密文跟前面的密文是有關的

不同 symmetric cipher 的 performance 比較：

Cipher	Key Length	Mbit/s
DES	56	36.95
3DES	112	13.32
AES	128	51.19
RC4 (stream cipher)	(choosable)	211.34

4.2.2. Random Number Generators (RNGs)



TRNGs

- based on physical random processes (擲骰子或硬幣這種)
- output stream s_i should have good statistical properties ($Pr(s_i = 0) = Pr(s_i = 1) = \frac{1}{2}$)
- output 不能被 predict 或 reproduce
- used to generate keys or nonces (used only-once values), etc.

PRNG

- generate sequence from initial seed value
- output stream has good statistic properties
- output can be reproduced and can be predicted (**bad cryptographic properties**)
- computed in recursive way :
 - $s_0 = \text{seed}$
 - $s_{i+1} = f(s_i, s_{i-1}, \dots, s_{i-t})$

e.g. Simple PRNG : **Linear Congruential Generator**

$S_0 = \text{seed}$

$S_{i+1} = AS_i + B \bmod m$

⇒ Assume A, B, S_0 are unknown and all 100 bits

300 bits of output are known (S_1, S_2, S_3)

⇒ 只要解聯立方程式就有 A, B 了

$S_2 = AS_1 + B(\bmod m)$

$S_3 = AS_2 + B(\bmod m)$

CSPRNG

- special PRNG with additional property
 - **output must be unpredictable**
 - given n consecutive bits of output s_i , the following output bits S_{n+1} can not be predicted (in polynomial time)
- need in cryptography, in particular for stream ciphers
- 除了 cryptography 幾乎沒有其他 application 需要 unpredictability (但很多需要 PRNG)

4.2.3. One-Time Pad (OTP)

Unconditionally Secure Cryptosystem : A cryptosystem is unconditionally secure if it can't be broken with *infinite* computational resources

One-Time Pad

Let the plaintext, ciphertext, and key consist of individual bits $x_i, y_i, k_i \in \{0, 1\}$

- encryption : $e_{k_i}(x_i) = x_i \oplus k_i$
- decryption : $d_{k_i}(y_i) = y_i \oplus k_i$

OTP is unconditionally secure iff the key k_i is used once :

$$\begin{aligned} y_0 &= x_0 \oplus k_0 \\ y_1 &= x_1 \oplus k_1 \\ &\vdots \end{aligned}$$

Every equation is a linear equation with two unknowns

⇒ for every y_i , $x_i = 0$ and $x_i = 1$ are equiprobable (等機率)

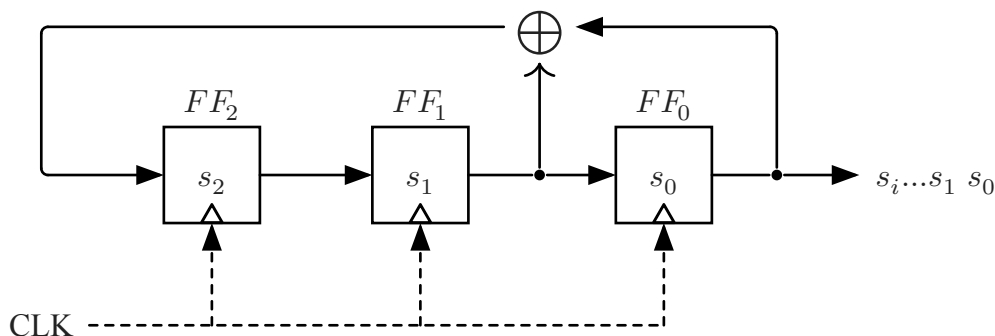
⇒ this is true iff $k_0, k_1, \dots$ independent (k_i 的生成是完全隨機的)

⇒ it can be shown that this systems can *provably* not be solved

缺點 : 很不切實際因為 key 跟 message 要一樣長

4.2.4. Linear Feedback Shift Registers (LFSRs)

LFSR of $m = 3$:

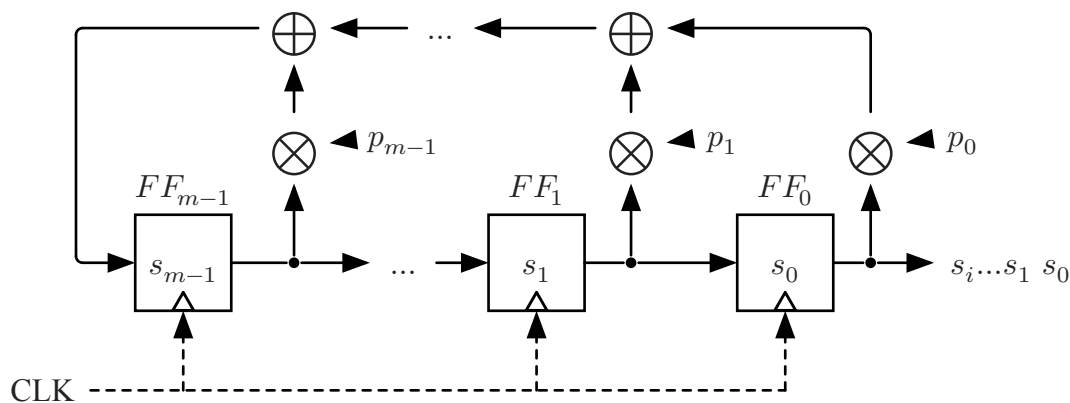


- Degree m given by the number of storage elements
- LFSR output described by recursive equation : $s_{i+3} = s_{i+1} + s_i \text{ mod } 2$
- max output length ($2^m - 1$) achieved only for certain feedback configurations (如下圖)

clk	0	1	2	3	4	5	6	7	8
FF_2	1	0	1	1	1	0	0	1	0
FF_1	0	1	0	1	1	1	0	0	1
$FF_0 = s_i$	0	0	1	0	1	1	1	0	0

會發現 $\text{clk} = 0$ 跟 $\text{clk} = 7$ 的時候是一樣的 $\Rightarrow$ 週期 = 7

All LFSRs :

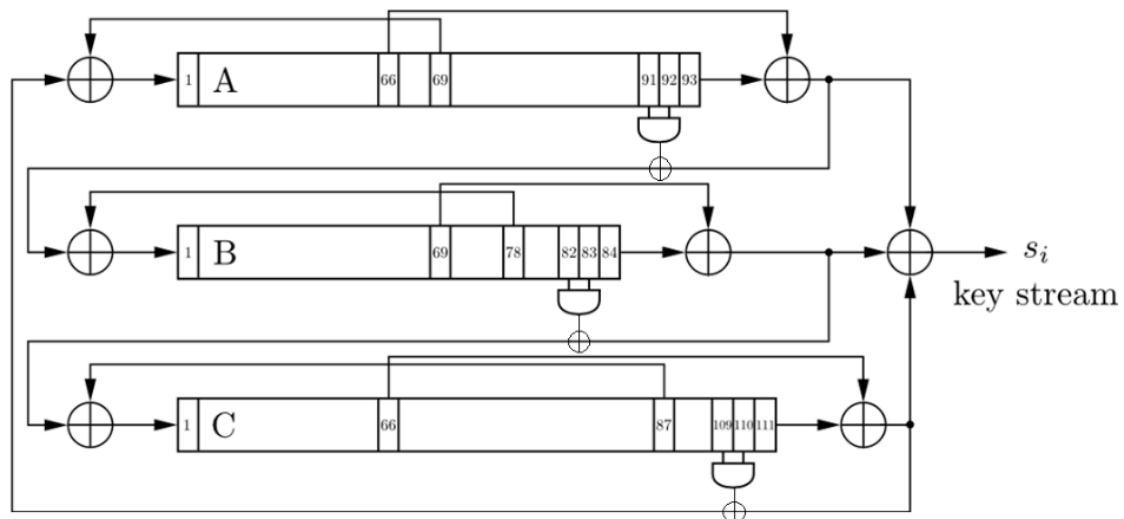


- $p_1 = 1$ 的時候是通路 $\Rightarrow$ 會上去 XOR
- output sequence repeats periodically (有週期)
- max period length of output : $2^m - 1$

$$P(x) = x^m + p_{m-1}x^{m-1} + \dots + p_1x + p_0$$

- single LFSRs generate highly predictable output
- if $2m$ output bits of an LFSR of degree m are known, the feedback coefficients p_i of the LFSR can be found by solving a system of linear equations
- because of this many stream ciphers use **combinations** of LFSRs

4.2.5. Trivium - A Modern Stream Cipher



	Register Length	Feedback Bit	Feedforward Bit 和結果做 XOR	AND Inputs AND 完 XOR
A	93	69	66	91, 92
B	84	78	69	82, 83
C	111	87	66	109, 110

small in hardware :

- 3 AND-Gates (no linearity)
- 7 XOR-Gates (4 with 3 inputs)
- 288 registers

initialization :

- load 80-bit key into A
- load 80-bit IV into B (IV 是 initialization vector, 是每次換的 nonce)
- set $C_{109}, C_{110}, C_{111} = 1$, others = 0

warm-up :

- clock cipher $4 \times 288 = 1152$ times without generating output

encryption :

- XOR sum of all three NLFSR outputs generates key stream s_i

design can be parallelized to produce up to 64 bits of output per clock cycle